Detailed Theory: Router Booting, Configuration Registers, Router IOS, Telnet, Hostname Resolution, and Debugging

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1. Router Booting Process

The Cisco router boot process is a series of steps the router follows when powered on or rebooted:

Boot Sequence Steps:

1. **POST (Power-On Self Test):**
   The router runs diagnostics on hardware (CPU, memory, interfaces).

2. **Bootstrap Program Execution:**
   The bootstrap is a small program in ROM that locates and loads the IOS.

3. **Loading IOS:**
   The bootstrap loads the IOS image from Flash memory (default). If not found, it tries TFTP or ROM IOS.

4.

Loading Configuration File:
The IOS loads the startup configuration file from NVRAM (non-volatile RAM). If no config file exists or is corrupted, the router enters setup mode.

5.

Router Initialization:
The router initializes interfaces and services and becomes operational.

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2. Configuration Register

- The **configuration register** controls the router's boot behavior.
- It is a 16-bit value (usually displayed in hexadecimal).
- The most common configuration register values:

Register Value	Description

0x2102	Default; boots from Flash and loads startup-config from NVRAM
0x2142	Ignore startup-config in NVRAM (used for password recovery)
0x2101	Boot IOS from ROM (ROMMON mode)

How to View and Change Configuration Register:

- View: show version
- Change:

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  Router(config)# config-register 0x2142 Router# reload

3. Router IOS

- Cisco IOS (Internetwork Operating System) is the router’s software image.
- Stored in Flash memory.
- Supports multiple protocols and features.
- Can be upgraded by copying newer IOS images via TFTP or USB.

4. Telnet

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Telnet is a protocol for remote CLI access over the network.

- Unsecured, transmits data (including passwords) in plaintext.

- Steps to enable Telnet on Cisco routers:

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Router(config)# line vty 0 4 Router(config-line)# password your_password Router(config-line)# login Router(config-line)# transport input telnet

- Telnet access is often replaced by SSH for security.

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5. Resolving Hostname

- Hostname resolution converts a hostname (like router1) to an IP address.
- On routers, hostname resolution can be done by:
  - **Local Hosts Table:** Manually configured with ip host commands.
  - **DNS:** Using the ip domain-lookup and ip name-server commands to query DNS servers.
- Commands:

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Router(config)# ip domain-lookup Router(config)# ip name-server 8.8.8.8 Router# ping hostname

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6. Debugging on Cisco Routers

- Debugging is used to display real-time information about router processes.
- Very powerful but can impact performance, so use carefully.

Common Debug Commands:

- debug ip packet – shows IP packets being processed
- debug interface – shows interface events
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debug arp – shows ARP requests and replies

- debug ssh – debug SSH sessions

- debug telnet – debug Telnet sessions

Stopping Debug:

- Use undebug all or no debug all.

Tips:

- Always monitor CPU usage during debugging.

- Use specific debug commands to minimize overhead.

Summary Table:

Topic	Key Points
Router Booting	POST, Bootstrap, IOS load, config load, initialization
Config Register	Controls boot behavior; 0x2102 default, 0x2142 ignore config
IOS	Stored in Flash; supports routing & services
Telnet	Remote CLI access; insecure; configure via line vty
Hostname Resolution	Use ip host or DNS servers; enable with ip domain-lookup
Debugging	Real-time process info; use carefully; stop with undebug-all